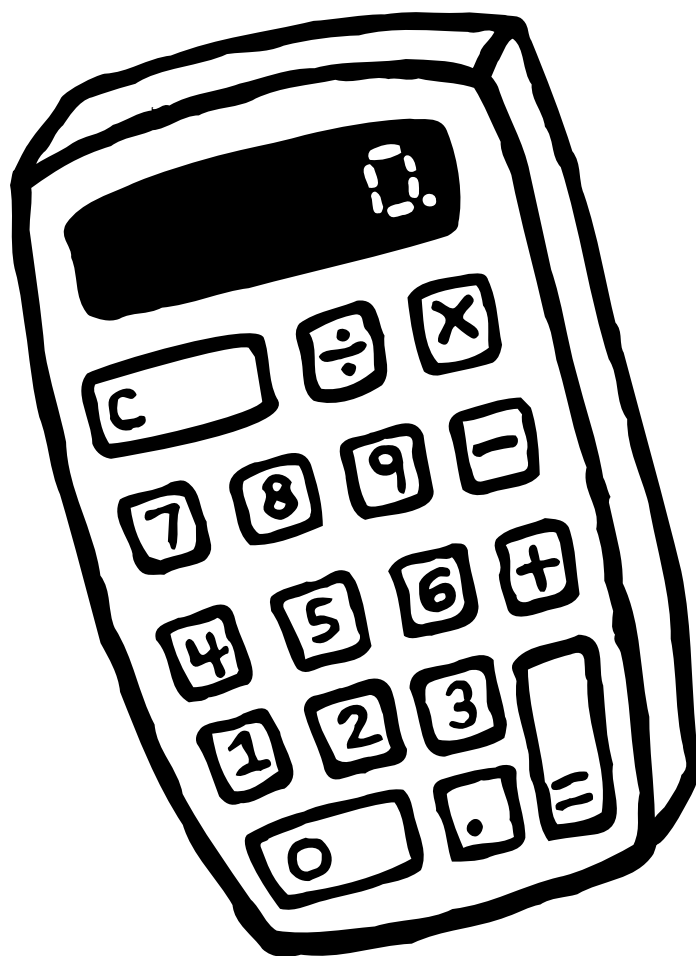




UNIVERSITY INTERSCHOLASTIC LEAGUE

# Mathematics

District • 2019



DO NOT TURN THIS PAGE UNTIL  
YOU ARE INSTRUCTED TO DO SO!

1. Evaluate:  $3 + 2 \times 5 - 3 + 0! \times 2^0 \div (1 + 9)$

- (A) 2.3                      (B) 10                      (C) 10.1                      (D) 10.2                      (E) 32

2. The local Country Bumpkin grocery is having a sale on all canned vegetables. Willie Byette needs to purchase 10 cans of vegetables. Which of the following is the least expensive deal, assuming all of the cans are of the same size and quality?

- (A) regular price at 59¢ each                      (B) \$2.75 for a 4-pack and get one free  
(C) 65¢ each less 15% off coupon                      (D) \$8.00 for a 12-pack  
(E) 3 for \$1.75

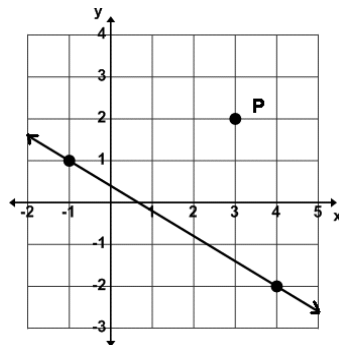
3. The sales tax on \$24.00 is \$1.98. What would the total cost including sales tax be on \$118.50? (nearest cent)

- (A) \$130.62                      (B) \$128.28                      (C) \$126.75                      (D) \$119.58                      (E) \$119.33

4. Let  $m = \{m, a, r, c, h\}$ ,  $M = \{m, o, n, d, a, y\}$ , and  $F = \{f, r, i, d, a, y\}$ . The number of elements in  $(m \cap M) \cup (m \cap F)$  is:

- (A) 1                      (B) 3                      (C) 4                      (D) 5                      (E) 6

5. Line  $m$  through point P is perpendicular to the line shown. Which of the following points lie on line  $m$  in quadrant III



- (A)  $(-3, -8)$                       (B)  $(-1, -5)$                       (C)  $(-1, -2)$                       (D)  $(-5, -2)$                       (E)  $(-3, -2)$

6. Let  $(5x - 1)^3 = ax^3 + bx^2 + cx + d$ . Find  $a + b + c + d$ .

- (A) 216                      (B) 184                      (C) 114                      (D) 65                      (E) 64

7. Which of the following is/are examples of a function?

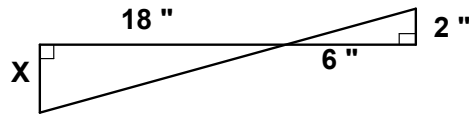
I.  $\{(2,3), (3,4), (4,5), (5,6)\}$     II.  $\{(1,1), (2,3), (3,5), (5,8)\}$     III.  $\{(-2, -1), (-1, 2), (0,0), (3,3)\}$

- (A) I only                      (B) II only                      (C) I & II                      (D) II & III                      (E) all of them

8. Ester Buney divided 4 dozen eggs among Huey, Duey, and Luey. She gave Luey three times as many eggs as Huey, and she gave Duey 3 more eggs than Huey. How many eggs did Duey get?

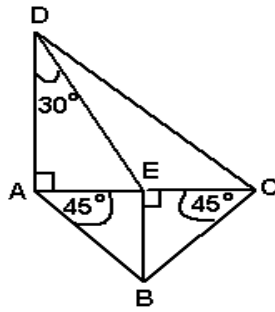
- (A) 16                      (B) 15                      (C) 13                      (D) 12                      (E) 9

9. Find  $x$ . (drawing is not to scale)



- (A) 9 "      (B) 6 "      (C) 1.5 "      (D)  $\frac{2}{3}$  "      (E) cannot be determined
10. Which of the following points of concurrency is on the right triangle vertex point, on the interior of an acute triangle, and on the exterior of an obtuse triangle?
- (A) centroid    (B) circumcenter    (C) incenter    (D) orthocenter    (E) none of these
11. A graph of a relation has a line of symmetry of  $x = 3$ . The graph passes through the point  $(-1, 7)$ . What is the  $x$ -coordinate of another point that must have a  $y$ -coordinate of 7?
- (A)  $-6$       (B)  $-5$       (C) 2      (D) 7      (E) 10
12. A right cylinder can of *Bugs* carrots has a lateral surface area of  $72 \text{ in}^2$  and a height of 4 in. What is the surface area of the base of the can of carrots? (nearest tenth)
- (A)  $32.8 \text{ in}^2$     (B)  $31.4 \text{ in}^2$     (C)  $28.3 \text{ in}^2$     (D)  $26.7 \text{ in}^2$     (E)  $25.8 \text{ in}^2$
13. Let  $3x - 2y = 4$  and  $x + 3y = 5$ . Find  $3x - 3y$ .
- (A) 0      (B)  $-17\frac{4}{7}$       (C) 3      (D)  $-5\frac{8}{11}$       (E) 6
14. Saul Would can cut a cord of wood in 2 hours. Tim Burr can cut a cord of wood in 1 hour and 20 minutes. How long would it take them to cut a cord if they work together?
- (A) 40 min    (B) 48 min    (C) 52 min    (D) 60 min    (E) 100 min
15. Let function  $f = \{(8, 5), (2, 8), (11, 4), (5, 6)\}$  and function  $g = \{(4, 5), (8, 0), (6, 3), (0, 8)\}$ . Find  $(f \circ g)(4) - (g \circ f)(5)$  ?
- (A)  $-1$       (B) 1      (C) 3      (D) 5      (E) 9
16. Determine the range of  $f(x) = 2 - 3\sin(5x + 7)$ .
- (A)  $[-1, 1]$     (B)  $[-5, -1]$     (C)  $[-3, 2]$     (D)  $[-2, 3]$     (E)  $[-1, 5]$
17. If  $f(x) = 3x^2 - 2x + 5$  and  $g(x) = 5x^2 - 3x + 2$ , then  $g(f(-1)) = ?$
- (A) 472      (B) 5      (C) 559      (D) 4      (E) 346
18.  $(3x^2 + kx + 2) \div (x + 5)$  has a remainder of 32. Find  $k$ .
- (A)  $-5$       (B)  $-3$       (C) 5      (D) 7      (E) 9

- 19. Find the perimeter of the quadrilateral ABED if  $AE = 20$  in. (nearest inch)**



- (A) 96 in      (B) 117 in      (C) 120 in      (D) 123 in      (E) 126 in

20. Find the sum of the coefficients of the  $x^4y^2$  term,  $x^3y^3$  term, and  $x^2y^4$  term in the expansion of  $(2x + y)^6$ .

- (A) 460                      (B) 520                      (C) 60                      (D) 180                      (E) 2,560

21. Let  $f''(x) = -6$ ,  $f'(2) = -1$ , and  $f(-1) = -6$ . Find  $f(0)$ .

- (A) 16                      (B)  $-1$                       (C) 8                      (D) 0                      (E)  $-13$

- 22. Max E. Mumm is fencing in a non-square rectangular area of 1,800 square feet. The cost of the fencing is \$7.00 per foot for the two short sides and \$5.00 per foot for the two long sides. What is the lowest possible cost for the fence?**

- (A) \$1004.00      (B) \$1018.23      (C) \$1020.00      (D) \$1030.00      (E) \$1180.00

23. Cash Droor has five \$20.00 bills, three \$10.00 bills, and two \$5.00 bills. If he selects two bills at random what is the probability that one is a \$20.00 bill and one is a \$10.00 bill?  
(nearest whole percent)

- (A) 6%**                      **(B) 11%**                      **(C) 17%**                      **(D) 33%**                      **(E) 67%**

24. Principal Rewler selected 5 teachers, 8 students, and 4 parents at his school to form an advisory committee. How many different 6-member committees can he form if 2 are teachers, 3 are students, and 1 is a parent?

- (A) 960                      (B) 1,120                      (C) 2,016                      (D) 2,240                      (E) 12,376

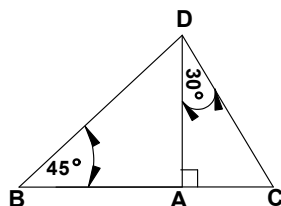
- 25. 113 is an emirp prime. How many emirp primes less than 113 exist?**

- (A) 7                      (B) 8                      (C) 9                      (D) 10                      (E) 11

26. Which of the following mathematicians proved that the number of elements of the infinite set of real numbers is greater than the number of elements of the infinite set of natural numbers?

- (A) Georg Cantor                      (B) Marin Mersenne                      (C) Franciscus Vieta  
(D) Leonard Euler                      (E) Aryabhata

27. Using dry volume measurements: 3 bushels — 1 peck + 2 gallon — 1 quarts = \_\_\_\_\_ pints.
- (A) 46                      (B) 92                      (C) 174                      (D) 190                      (E) 200
28. The least common multiple of 315, 525, and 735 minus the greatest common factor of 315, 525, and 735 is?
- (A) 1,470                      (B) 2,205                      (C) 1,675                      (D) 10,920                      (E) 11,130
29. Let  $9x^2 - 9x - 4 = (ax + b)(cx + d)$ . Find  $a + b - c + d$ .
- (A) -5                      (B) -4                      (C) -3                      (D) 3                      (E) 10
30. Find  $f(2) + f(3) + f(-5)$  if  $f(x) = \begin{cases} 2x - 1 & \text{if } x < 0 \\ 3x & \text{if } 0 \leq x \leq 3 \\ 1 - 4x & \text{if } x > 3 \end{cases}$
- (A) 14                      (B) 4                      (C) 0                      (D) -11                      (E) -16
31. The area of  $\triangle ABD$  contains how many more square feet than the area of  $\triangle ACD$  if  $AC = 3$  ft. (nearest tenth)

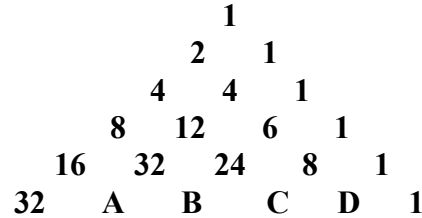


- (A) 8.4 ft<sup>2</sup>                      (B) 1.8 ft<sup>2</sup>                      (C) 5.7 ft<sup>2</sup>                      (D) 7.8 ft<sup>2</sup>                      (E) 13.5 ft<sup>2</sup>
32. An obtuse angle is bisected. One of the acute angles created by bisecting the original obtuse angle is trisected. One of the smallest acute angles is bisected. The measure of the smallest angle is 12°. What was the angle measure of the original obtuse angle?
- (A) 144°                      (B) 132°                      (C) 120°                      (D) 112°                      (E) 108°
33. Let  $A = \begin{bmatrix} -1 & 4 \\ 3 & 6 \end{bmatrix}$  and  $B = \begin{bmatrix} 2 & 7 \\ 5 & -8 \end{bmatrix}$ . If  $AB = \begin{bmatrix} d & e \\ f & g \end{bmatrix}$  then  $ef = ?$
- (A) -1,404                      (B) -486                      (C) 648                      (D) 918                      (E) 1,053
34. If  $a_1 = -2$ ,  $a_2 = 0$ ,  $a_3 = 2$  and  $a_n = a_{n-2} - a_{n-1} + a_{n-3}$ , where  $n \geq 4$ , then  $a_7$  equals:
- (A) 10                      (B) 2                      (C) 0                      (D) -4                      (E) -8
35. The expression  $\cos \theta \div (1 - \sin \theta) - \tan \theta$  is equivalent to:
- (A)  $\sec \theta$                       (B)  $\frac{\cos \theta}{1 - \sin \theta}$                       (C)  $\cot \theta$                       (D)  $\frac{1}{\csc \theta}$                       (E)  $\frac{1}{\sec \theta}$

36. Saul T. Water anchored a buoy 200 feet from his pier on a bearing of  $320^\circ$ . He places a second buoy 150 feet from his pier on a bearing less than  $90^\circ$ . What bearing would the second buoy have to be placed in order for the buoys to be 300 feet apart? (nearest degree)

(A)  $84^\circ$       (B)  $77^\circ$       (C)  $67^\circ$       (D)  $50^\circ$       (E)  $40^\circ$

37. Find  $A + B + C + D$  if the triangular pattern shown below continues.



(A) 248      (B) 236      (C) 224      (D) 210      (E) 202

38. Les Cash invested his inheritance in a mutual fund. At the end of the first year the fund decreased 7%. The fund increased 2% at the end of the second year, then decreased 5% at the end year three. What was Les' average rate of return over the four year period if the fund increased 3% the fourth year? (nearest tenth)

(A)  $-1.8\%$       (B)  $-1.0\%$       (C)  $+1.0\%$       (D)  $+2.2\%$       (E)  $+3.4\%$

39. Find the slope of the line tangent to  $f(x) = 2x^2 - 13x + 5$  at  $x = 3$ .

(A) 4      (B) 3      (C) 0      (D)  $-1$       (E)  $-4$

40. Let  $f(x) = \frac{x^2 - 5}{x + 1}$ . How many of the following statements are always true?

I.  $\lim_{x \rightarrow -1^+} f(x)$  exists

II.  $\lim_{x \rightarrow -1^-} f(x)$  exists

III.  $f(x)$  is continuous

IV.  $f(x)$  is differentiable at  $x = 1$

(A) 0      (B) 1      (C) 2      (D) 3      (E) 4

41. Willie Chuzrite selects one number from each of the sets  $\{1, 2, 3, 5, 7\}$  and  $\{0, 4, 6, 8, 9\}$ . What is the probability he selects two numbers totaling a sum greater than 10 given that a prime number is chosen from the first set? (nearest whole percent)

(A) 33%      (B) 40%      (C) 50%      (D) 60%      (E) 67%

42. Roland Bones tosses a pair of dice. What are the odds the sum of the dots (pips) on the top faces is greater than seven?

(A) 5:7      (B) 5:11      (C) 1:7      (D) 7:12      (E) 5:12

43. 1,088 is the  $k^{\text{th}}$  term of the sequence 0, 3, 8, 15, 24, 35, 48, ... . Find  $k$ .

(A) 23      (B) 27      (C) 31      (D) 33      (E) 37

44.  $(642_7 - 135_7) \times 6_7 = \underline{\hspace{2cm}}_7$ .
- (A) 4303      (B) 4233      (C) 3313      (D) 3042      (E) 3024
45. If  $\sqrt[5]{x} \sqrt[3]{x} \sqrt{x} = \sqrt[n]{x^k}$ , where  $k$  and  $n$  are relatively prime, then  $k + n = ?$
- (A) 24      (B) 22      (C) 19      (D) 15      (E) 13
46. Let  $(-1 + \sqrt{3}i)^{-8} = (a + bi)$ . Find  $a + b$ . (nearest ten-thousandth)
- (A) .0082      (B) .0054      (C) .0032      (D) .0018      (E) .0014
47. An odd number  $N$  exists such that  $500 < N < 600$  and  $N$  is divisible by both 7 and 11. The sum of the digits of the number  $N$  is?
- (A) 18      (B) 17      (C) 16      (D) 15      (E) 14
48.  $\int \sin(x) \cos(x) dx = ?$
- (A)  $-\cos(x) \sin(x) + C$       (B)  $\frac{\sin^2(x)}{4} + C$       (C)  $-\frac{\cos(2x)}{4} + C$   
 (D)  $\cos^2(x) + C$       (E)  $-\frac{\cos^2(2x)}{4} + C$
49. The *Mixed Fruit* shop sells fruit bowls consisting of 6 pieces of fruit. The shop has oranges, bananas, apples, grapes, peaches, pears, and plums. How many different fruit bowls can they make to sell?
- (A) 1,716      (B) 132      (C) 5,040      (D) 720      (E) 924
50. Suppose the radius of a right circular cylinder is decreasing at the rate of 4 feet per minute, while the height is increasing at the rate of 2 feet per minute. Find the rate of change in the volume when the radius is 2 feet and the height is 6 feet. (nearest  $\frac{\text{ft}^3}{\text{min}}$ )
- (A)  $-75 \frac{\text{ft}^3}{\text{min}}$       (B)  $-101 \frac{\text{ft}^3}{\text{min}}$       (C)  $-125 \frac{\text{ft}^3}{\text{min}}$       (D)  $-151 \frac{\text{ft}^3}{\text{min}}$       (E)  $-276 \frac{\text{ft}^3}{\text{min}}$
51. The sequence 1, 5, 6, 11, 17, ... is a Fibonacci characteristic sequence, where 1 is the first term. Find the sum of the first 11 terms.
- (A) 789      (B) 798      (C) 799      (D) 804      (E) 880
52. Find the area of the region bounded by the curve  $y = x^3$ ,  $x = -2$ , and  $x = 1$ .
- (A) 4.75 sq. units      (B) 4.25 sq. units      (C) 4.125 sq. units      (D) 3.875 sq. units      (E) 3.75 sq. units
53. The arithmetic mean of 3, 25, and 30, is 19.3333. Which of the following trapezoidal means of 3, 25, and 30 would be 7.4 if rounded to the nearest tenth?
- (A) Geometric      (B) Heronian      (C) Centroidal      (D) Contraharmonic      (E) Harmonic

54. Find the sum of the  $x$ -values in  $\left\{x \mid 5\cos^2(x) = 4 - 3\sin^2(x), x \in [0, \frac{3\pi}{2}] \right\}$ . (nearest hundredth)
- (A) 6.44            (B) 7.07            (C) 7.23            (D) 7.854            (E) 8.02
55. The harmonic mean of the roots of  $x^4 - 13x^3 + 56x^2 - 92x + 48 = 0$  is ? (nearest tenth)
- (A) 3.4            (B) 3.3            (C) 2.6            (D) 2.3            (E) 2.1
56. The medians of a triangle are concurrent. The length of one of the medians is 24 cm. What is the length of the segment joining the midpoint of a side of the triangle to the point of concurrency?
- (A) 16 cm            (B) 12 cm            (C) 8 cm            (D) 6 cm            (E) 4 cm
57. Let  $\frac{a-b}{3} = 1.5$ ,  $\frac{a-c}{2} = 3$ , and  $\frac{b+c}{5} = \frac{2}{5}$ . Find the arithmetic mean of  $a$ ,  $b$ , and  $c$ .
- (A)  $2\frac{3}{4}$             (B)  $2\frac{7}{12}$             (C)  $1\frac{1}{4}$             (D)  $1\frac{5}{12}$             (E)  $1\frac{19}{30}$
58.  $32P3Q2019 \div 9$  has a remainder of 7. Find the least value of  $P + Q$ .
- (A) 9            (B) 8            (C) 7            (D) 6            (E) 5
59. Expand  $10^B \div (10^{(2B)} - 10^B - 1)$  for  $B = 4$ . What is the 24<sup>th</sup> digit after the decimal place?
- (A) 0            (B) 1            (C) 3            (D) 5            (E) 8
60. How many distinct 5-letter code words can be made from the letters in the words HYPATIA?
- (A) 2,520            (B) 1,320            (C) 1,020            (D) 720            (E) 120

**DO NOT DISTRIBUTE TO STUDENTS BEFORE OR DURING THE CONTEST**

**University Interscholastic League  
MATHEMATICS CONTEST  
HS • District • 2019  
Answer Key**

|              |                              |              |
|--------------|------------------------------|--------------|
| <b>1. C</b>  | <b>21. C</b>                 | <b>41. C</b> |
| <b>2. B</b>  | <b>22. No Correct Answer</b> | <b>42. A</b> |
| <b>3. B</b>  | <b>23. D</b>                 | <b>43. D</b> |
| <b>4. B</b>  | <b>24. D</b>                 | <b>44. B</b> |
| <b>5. A</b>  | <b>25. C</b>                 | <b>45. E</b> |
| <b>6. E</b>  | <b>26. A</b>                 | <b>46. E</b> |
| <b>7. E</b>  | <b>27. D</b>                 | <b>47. B</b> |
| <b>8. D</b>  | <b>28. D</b>                 | <b>48. C</b> |
| <b>9. B</b>  | <b>29. C</b>                 | <b>49. E</b> |
| <b>10. D</b> | <b>30. B</b>                 | <b>50. E</b> |
| <b>11. D</b> | <b>31. C</b>                 | <b>51. D</b> |
| <b>12. E</b> | <b>32. A</b>                 | <b>52. B</b> |
| <b>13. C</b> | <b>33. A</b>                 | <b>53. E</b> |
| <b>14. B</b> | <b>34. A</b>                 | <b>54. B</b> |
| <b>15. C</b> | <b>35. A</b>                 | <b>55. E</b> |
| <b>16. E</b> | <b>36. B</b>                 | <b>56. C</b> |
| <b>17. A</b> | <b>37. D</b>                 | <b>57. A</b> |
| <b>18. E</b> | <b>38. A</b>                 | <b>58. E</b> |
| <b>19. D</b> | <b>39. D</b>                 | <b>59. E</b> |
| <b>20. A</b> | <b>40. B</b>                 | <b>60. B</b> |

University Interscholastic League  
MATHEMATICS CONTEST

WRITE ALL ANSWERS WITH  
CAPITAL LETTERS

|       |       |          |
|-------|-------|----------|
| Final | _____ | _____    |
| 2nd   | _____ | _____    |
| 1st   | _____ | _____    |
| Score |       | Initials |

|                    |                  |
|--------------------|------------------|
| Contestant # _____ | Conference _____ |
|--------------------|------------------|

- |           |           |           |
|-----------|-----------|-----------|
| 1. _____  | 21. _____ | 41. _____ |
| 2. _____  | 22. _____ | 42. _____ |
| 3. _____  | 23. _____ | 43. _____ |
| 4. _____  | 24. _____ | 44. _____ |
| 5. _____  | 25. _____ | 45. _____ |
| 6. _____  | 26. _____ | 46. _____ |
| 7. _____  | 27. _____ | 47. _____ |
| 8. _____  | 28. _____ | 48. _____ |
| 9. _____  | 29. _____ | 49. _____ |
| 10. _____ | 30. _____ | 50. _____ |
| 11. _____ | 31. _____ | 51. _____ |
| 12. _____ | 32. _____ | 52. _____ |
| 13. _____ | 33. _____ | 53. _____ |
| 14. _____ | 34. _____ | 54. _____ |
| 15. _____ | 35. _____ | 55. _____ |
| 16. _____ | 36. _____ | 56. _____ |
| 17. _____ | 37. _____ | 57. _____ |
| 18. _____ | 38. _____ | 58. _____ |
| 19. _____ | 39. _____ | 59. _____ |
| 20. _____ | 40. _____ | 60. _____ |